Systems Engineering for Unmanned Aerial Vehicles

Part II – Tutorial 1

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Tutorial questions

1. Hexacopter modeling and control
2. VTOL tailsitter modeling
3. Quadcopter and FixedWing Simulink modeling
4. Airborne Wind Energy simulator

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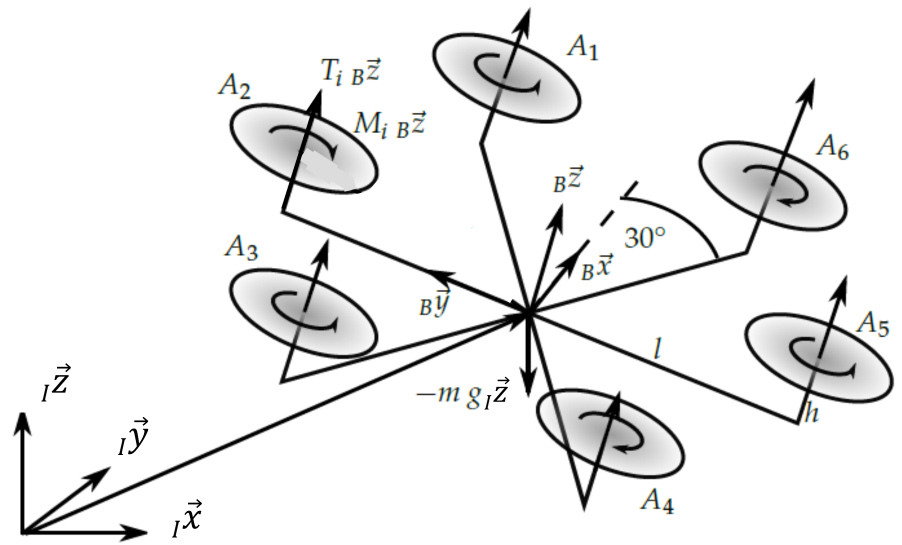
Tutorial question 1 – hexacopter modeling and control

1. Given the dynamics of a hexacopter shown below, with state vector: $\boldsymbol{X} = [{}_I\boldsymbol{x} \;\; {}_B\boldsymbol{v} \;\; \boldsymbol{X_r} \;\; \boldsymbol{\omega}]^T$, linearize the dynamics around hover assuming zero yaw angle ψ.

$$\dot{X} = \begin{bmatrix} {}_I\dot{\boldsymbol{x}} \\ {}_B\dot{\boldsymbol{v}} \\ \dot{\boldsymbol{X}}_r \\ \dot{\boldsymbol{\omega}} \end{bmatrix} \quad \begin{aligned} &\rightarrow {}_I\dot{\boldsymbol{x}} = \boldsymbol{C}_{IB} \cdot {}_B\boldsymbol{v} \\ &\rightarrow {}_B\dot{\boldsymbol{v}} = \frac{1}{m}\begin{bmatrix} 0 \\ 0 \\ U_1 \end{bmatrix} - \boldsymbol{C}_{IB}^T \begin{bmatrix} 0 \\ 0 \\ g \end{bmatrix} - \boldsymbol{\omega} \times {}_B\boldsymbol{v} \\ &\rightarrow \dot{\boldsymbol{X}}_r = \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = E_r^{-1}\,{}_B\boldsymbol{\omega} \\ &\rightarrow \dot{\boldsymbol{\omega}} = \boldsymbol{I}^{-1}\left[\begin{bmatrix} U_2 \\ U_3 \\ U_4 \end{bmatrix} - \boldsymbol{\omega} \times \boldsymbol{I}\boldsymbol{\omega}\right] \end{aligned}$$



$$E_r = \begin{bmatrix} 1 & 0 & -\mathrm{s}\,\theta \\ 0 & \mathrm{c}\,\phi & \mathrm{s}\,\phi\,\mathrm{c}\,\theta \\ 0 & -\mathrm{s}\,\phi & \mathrm{c}\,\phi\,\mathrm{c}\,\theta \end{bmatrix} \qquad C_{IB} = \begin{bmatrix} \mathrm{c}\psi\,\mathrm{c}\,\theta & \mathrm{c}\,\psi\,\mathrm{s}\,\theta\,\mathrm{s}\,\phi - \mathrm{s}\,\psi\,\mathrm{c}\,\phi & \mathrm{c}\,\psi\,\mathrm{s}\,\theta\,\mathrm{c}\,\phi + \mathrm{s}\,\psi\,\mathrm{s}\,\phi \\ \mathrm{s}\,\psi\,\mathrm{c}\,\theta & \mathrm{s}\,\psi\,\mathrm{s}\,\theta\,\mathrm{s}\,\phi + \mathrm{c}\,\psi\,\mathrm{c}\,\phi & \mathrm{s}\,\psi\,\mathrm{s}\,\theta\,\mathrm{c}\,\phi - \mathrm{c}\,\psi\,\mathrm{s}\,\phi \\ -\mathrm{s}\,\theta & \mathrm{c}\,\theta\,\mathrm{s}\,\phi & \mathrm{c}\,\theta\,\mathrm{c}\,\phi \end{bmatrix}$$

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$$\dot{v}_B = \begin{bmatrix} \dot{u} \\ \dot{v} \\ \dot{w} \end{bmatrix} = \frac{1}{m}\begin{bmatrix} 0 \\ 0 \\ U_1 \end{bmatrix} - C_{I,B}^{T}\begin{bmatrix} 0 \\ 0 \\ g \end{bmatrix} - \begin{bmatrix} p \\ q \\ r \end{bmatrix} \times \begin{bmatrix} u \\ v \\ w \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ \frac{U_1}{m} \end{bmatrix} - \begin{bmatrix} -\sin\theta g \\ \cos\theta \sin\phi g \\ \cos\theta \cos\phi g \end{bmatrix} - \begin{vmatrix} i & j & k \\ p & q & r \\ u & v & w \end{vmatrix}$$

$$= \begin{bmatrix} \sin\theta g - qw + rv \\ -\cos\theta \sin\phi g - ru + pw \\ \frac{U_1}{m} - \cos\theta\cos\phi g - pv + qu \end{bmatrix}$$

$$\dot{\omega} = \begin{bmatrix} \dot{p} \\ \dot{q} \\ \dot{r} \end{bmatrix} = \begin{bmatrix} I_{xx} & 0 & 0 \\ 0 & I_{yy} & 0 \\ 0 & 0 & I_{zz} \end{bmatrix}^{-1} \left(\begin{bmatrix} U_2 \\ U_3 \\ U_4 \end{bmatrix} - \begin{bmatrix} p \\ q \\ r \end{bmatrix} \times \begin{bmatrix} I_{xx} & 0 & 0 \\ 0 & I_{yy} & 0 \\ 0 & 0 & I_{zz} \end{bmatrix} \begin{bmatrix} p \\ q \\ r \end{bmatrix} \right)$$

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$$= \frac{1}{I_{xx} I_{yy} I_{zz}} \begin{bmatrix} I_{yy} I_{zz} & 0 & 0 \\ 0 & I_{xx} I_{zz} & 0 \\ 0 & 0 & I_{xx} I_{yy} \end{bmatrix} \begin{bmatrix} U_2 - (I_{zz} - I_{yy}) qr \\ U_3 - (I_{xx} - I_{zz}) pr \\ U_4 - (I_{yy} - I_{xx}) pq \end{bmatrix}$$

$$= \begin{bmatrix} \frac{U_2}{I_{xx}} + \frac{I_{yy} - I_{zz}}{I_{xx}} qr \\ \frac{U_3}{I_{yy}} + \frac{I_{zz} - I_{xx}}{I_{yy}} pr \\ \frac{U_4}{I_{zz}} + \frac{I_{xx} - I_{yy}}{I_{zz}} pq \end{bmatrix}$$

Around hover. $[u_0, v_0, w_0, \phi_0, \theta_0, \psi_0, p_0, q_0, r_0] = 0$.

applying the small perturbation:

$u(t) = u_0 + \Delta u = \Delta u$. same for v, w, p, q, r;

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$\sin\theta(t) = \sin(\theta_0 + \Delta\theta) \approx \sin\theta_0 + \Delta\theta\cos\theta_0 \approx \Delta\theta$, same for ϕ, ψ;

$\cos\theta(t) = \cos(\theta_0 + \Delta\theta) \approx \cos\theta_0 - \Delta\theta\sin\theta_0 \approx 1$, same for ϕ, ψ;

and neglecting all term equal and above 2nd order, such as $\Delta p \Delta r$,

yields:

$$\Delta \dot{v}_B = \begin{bmatrix} \Delta\dot{u} \\ \Delta\dot{v} \\ \Delta\dot{w} \end{bmatrix} = \begin{bmatrix} \Delta\theta g \\ -\Delta\phi g \\ \frac{U_1}{m} - g \end{bmatrix}$$

$$\Delta \dot{\omega} = \begin{bmatrix} \Delta\dot{p} \\ \Delta\dot{q} \\ \Delta\dot{r} \end{bmatrix} = \begin{bmatrix} \frac{U_2}{I_{xx}} \\ \frac{U_3}{I_{yy}} \\ \frac{U_4}{I_{zz}} \end{bmatrix}$$

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$$\Delta \dot{X}_r = \begin{bmatrix} \Delta\dot{\phi} \\ \Delta\dot{\theta} \\ \Delta\dot{\psi} \end{bmatrix} = \begin{bmatrix} 1 & 0 & -\Delta\theta \\ 0 & 1 & \Delta\phi \\ 0 & -\Delta\phi & 1 \end{bmatrix}^{-1} \begin{bmatrix} \Delta p \\ \Delta q \\ \Delta r \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \Delta p \\ \Delta q \\ \Delta r \end{bmatrix} = \begin{bmatrix} \Delta p \\ \Delta q \\ \Delta r \end{bmatrix}$$

$$\Delta \dot{x}_I = \begin{bmatrix} \Delta\dot{x}_I \\ \Delta\dot{y}_I \\ \Delta\dot{z}_I \end{bmatrix} = \begin{bmatrix} 1 & -\Delta\psi & \Delta\theta \\ \Delta\theta & 1 & -\Delta\phi \\ -\Delta\theta & \Delta\phi & 1 \end{bmatrix} \begin{bmatrix} \Delta u \\ \Delta v \\ \Delta w \end{bmatrix}$$

$$= \begin{bmatrix} \Delta u \\ \Delta v \\ \Delta w \end{bmatrix}$$

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Tutorial question 1 – hexacopter modeling and control

2. Draw a typical hierarchical control structure of a hexacopter.
3. Write the linear mapping $\mathbf{A}$ between the square of rotors angular velocity $\omega_{p,i}^2$ and the total forces and torques acting on the platform.
    1. $[U_1 \;\; U_2 \;\; U_3 \;\; U_4]^T = \mathbf{A}\left[\omega_{p,1}^2 \;\; \dots \omega_{p,6}^2\right]^T$
4. Write the control inputs U_2, U_3, U_4 as a function of measured attitude ϕ, θ, ψ, angular velocity ω and desired attitude ϕ_d, θ_d, ψ_d, using the standard PD control approach.
5. Design a velocity controller (e.g. P-controller) that outputs desired attitude ϕ_d, θ_d and total thrust U_1 as a function of measured velocity ${}_B\boldsymbol{v}$ and desired velocity ${}_B\boldsymbol{v}_d$.

2. see solution

3. for $T_i = b_i \omega_{p,i}^2$, where $b = \rho C_T A R^2$,

and $Q_i = d_i \omega_{p,i}^2$, where $d = \rho C_Q A R^3$,

$$\therefore \begin{bmatrix} U_1 \\ U_2 \\ U_3 \\ U_4 \end{bmatrix} = \underbrace{\begin{bmatrix} b_1 & b_2 & b_3 & b_4 & b_5 & b_6 \\ \frac{1}{2}b_1 l & b_2 l & \frac{1}{2}b_3 l & -\frac{1}{2}b_5 l & -b_5 l & -\frac{1}{2}b_6 l \\ \frac{\sqrt{3}}{2}b_1 l & 0 & -\frac{\sqrt{3}}{2}b_3 l & -\frac{\sqrt{3}}{2}b_5 l & 0 & \frac{\sqrt{3}}{2}b_6 l \\ d_1 & -d_2 & d_3 & -d_4 & d_5 & -d_6 \end{bmatrix}}_{\text{reverse sign of } T \text{ and } \psi \text{ since } z\text{-axis defined here pointed upward}} \begin{bmatrix} \omega_1^2 \\ \omega_2^2 \\ \omega_3^2 \\ \omega_4^2 \\ \omega_5^2 \\ \omega_6^2 \end{bmatrix}$$

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4. $$U_2 = k_p(\phi_d - \phi) - k_d\dot{\phi}.$$

$$U_3 = k_p(\theta_d - \theta) - k_d\dot{\theta}.$$

$$U_4 = k_p(\psi_d - \psi) - k_p\dot{\psi}.$$

where assume that $\dot{\theta}_d = \dot{\phi}_d = \dot{\psi}_d = 0$

5. $$\phi_d = -k_p(v_d - v)$$

$$\theta_d = k_p(u_d - u)$$

$$U_1 = k_p(w - w_d) + mg$$

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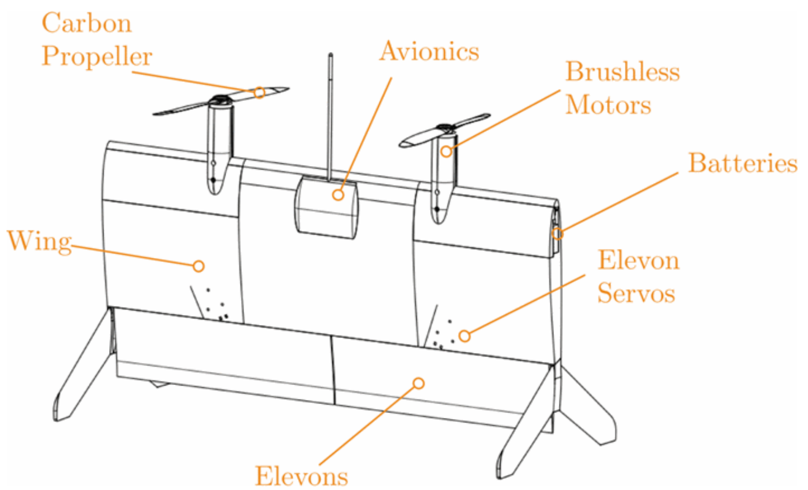
Tutorial question 2 – VTOL tailsitter UAV modeling

VTOL tailsitter UAVs combine the advantages of both fixed wing and rotary wing systems.

- How can the VTOL tailsitter UAV control its position and its yaw?
- Derive the forces and moments acting on the UAV in hover.



Verling et al 2016 – Full Attitude Control of a VTOL tailsitter UAV

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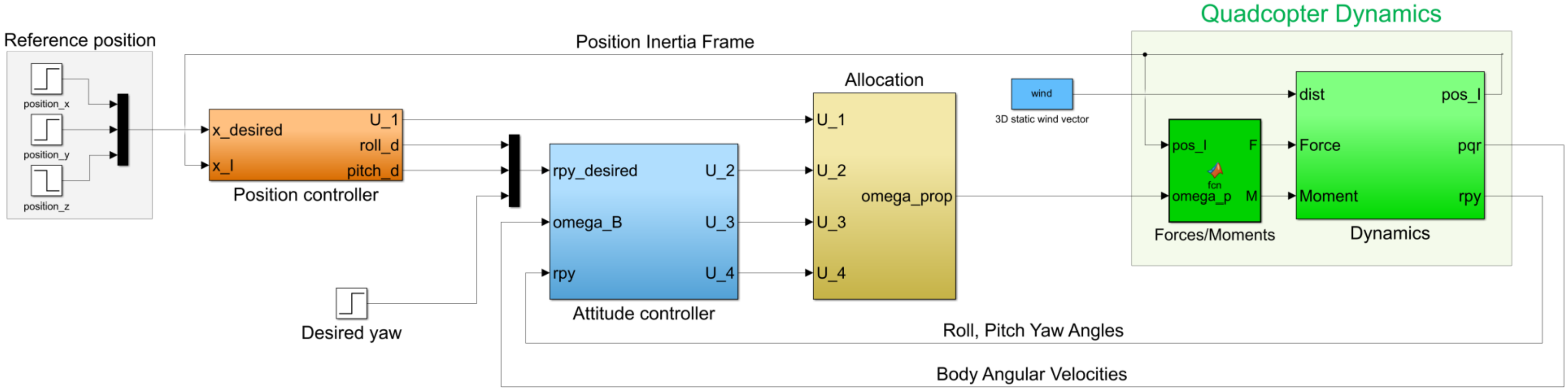
Tutorial question 3 – Quadcopter and fixed-wing simulator

Explore the matlab/simulink quadcopter and fixed-wing UAV simulator introduced in lecture 4 and 6

- **Quadcopter**:
  - Introduce a trajectory following controller similar to the fixed wing controller
- **Fixed-wing UAV**
  - Build a lift+cruise model by combining the quadcopter and fixed wing forces & moments function



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Tutorial question 4 – AWE matlab simulator

Explore the matlab AWE simulator

- https://github.com/awegroup/MegAWES
- 6DOF rigid body wind energy UAV dynamics
- Flight controller

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